SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

192511348

I B.VIJAY KARTHIKKAYAN *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

15/20

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CO1 AT3

AI

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CSE

1) Smart home Automation using intelligent.

Introduction

a smart home automation system controls appliances such as lights, fans, air conditioners heaters..

The system receives real time input from sensors like motion detector, temperature sensor camera and clock.

Types of Reflex agents:

1. Simple reflex agent:

Makes decision only based on the current percept using predefined if - Then rule

eg: if motion detected → turn on light.

if no motion detected → turn off light.

advantages: * very fast.

* easy to design.

* low computational cost.

2) Model Based Agent:

Maintain a internal model memory of the environment and keeps track of prev stat.

eg: * bedroom light is already on.

* AC is running.

advantages:

* Makes better decision

* uses memory

3) Goal Based agent:

agent select action that help achieve a predefined goal.

goals: * Maintain temperature at 24°C

* Minimize electricity usage.

4) Utility based agent,

agent choose the action that provides the highest overall benefits.

working: Temperature = 27°C

option 1: lower AC option 2: Open window + fan

conclusion all through each intelligent agent has unique strength it ensures max comfort minimum energy consumption.

deepening search (IDS)

This combines the advantages of DFS ~~&~~ BFS

eg DEPT 0

S

Depth - 1

S
A
B

DEPT 2

S
A
B
C
D

(until goal is reached)

advantages -

* low memory usage.
* complete.
* suitable when goal depth is unknown.

Real world applications

* rescue robots.
* indoor service robot
* delivery robots.

Conclusion.

UCS is ideal for finding the best cost paths in weighted environment and IDS is memory efficient for depth.

2. Robot navigation

Introduction : a robot is placed inside an unknown maze must search for a path from the start position to exit.

* uniform cost search
* iterative deepening search.

uniform cost search (UCS)

expands the node with the lowest cummulative path cost

$g(n)$ = Actual path cost from start to end.

working - different path have different cost

$S \rightarrow A \rightarrow G$ = cost 8

$S \rightarrow B \rightarrow C \rightarrow G$ = cost 5

UCS Selects

$S \rightarrow B \rightarrow C \rightarrow G$ because cost is minimum.

advantages

* find optimal path
* Complete algorithm.